Impacts of an Informal Astronomy Program on Science Voice

Implementing Skynet Junior Scholars

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Background

The Skynet Junior Scholars (SJS) program, designed by the University of Chicago and North Carolina State University, aims to engage participants in research by providing tools to conduct scientific explorations of the universe. Participants experience hands-on STEM based activities including the use of a global network of robotic telescopes to take and analyze their own astronomical images.

Purpose and Goals

- Explore the impacts of an informal astronomy curriculum on the science voice of participants.
- Bolster opportunities for underrepresented and underserved students in STEM. This is our target group.
- Maximize the learning affordances present in SJS.
- Continue to grow the program through collaboration with the Bell Museum and Urban 4-H

Urban 4-H

Questions

The primary research question is: *In what ways does SJS allow participants to demonstrate science voice?*

Related sub questions that may also be explored:

1. Are SJS impacts on voice the same for all participants?
2. What are the indicators that allow participants to demonstrate unique science voice?
3. What key experiences are involved with development of science voice?

Methods

Mixed methods will be applied. Qualitative components include a single embedded case design contextualized in an informal astronomy program with multiple units of analysis (Yin, 2014). The constructivist iterative sequence of learning and relearning by students and teachers also relates this to a design based theoretical framework. This study is framed by Gee's claims relating academic social language and science learning (Yerrick, & Roth, 2005).

Evaluate Four Components to Indicate Developing Science Voice

1. Active participation in an immersive experience of multiple models integrated with situational practice that intentionally imbues appropriation of scientific language.

Participant science voice will be evaluated using:

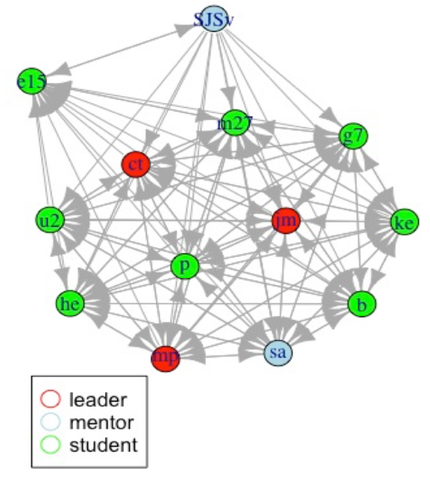
- written and verbal responses to prompts
- Interviews with participant and parents
- participant created astronomical images
- instructor observations
- earned badges
- class participation

2. Rich and redundant interaction with a learning community of peer and professional mentors who coach learners to match form to function as they practice using scientific language.

- Initial inductive coding then deductive coding.
- Apply SNA principles to analyze the forum interactions by comparing impacts of leaders, mentors, and peers.
- Pre and post-tests will be analyzed with paired t tests and ANOVA to look for differences between groups.

3. Express a science identity:
a. establish and progress toward scientific goals.
b. interact positively with community of practice.
c. bridge current cultural identity with science culture.

SJS 'Scavenger Hunt' Forum Discussion



size	density	reciprocity	diameter	network size
13	0.865	0.84	2	139

betweeness centrality	degree centrality	closeness centrality	eigenvector centrality
0.295	0.17	0.389	0.22

4. Demonstrate self-efficacy by performing authentic scientific research and communicating personally meaningful results in multiple modes to a community of scientific practice for examination, learning, and critique.

"The recitation becomes the social clearing-house, where experiences and ideas are exchanged and subjected to criticism, where misconceptions are corrected, and new lines of thought and inquiry are set up" (Dewey, 1932, p.49).

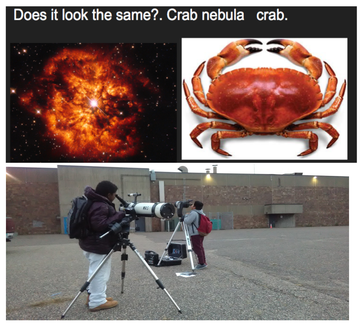
Preliminary Findings Spring 2017 Pilot Case Study

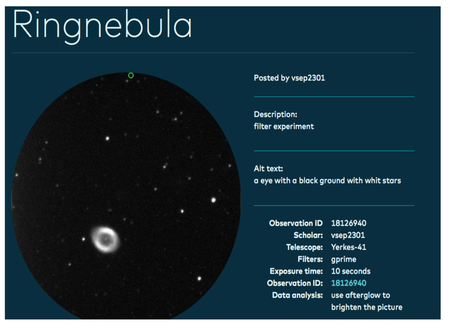
- SJS appears to facilitate the expression of science voice for middle school students from groups traditionally underrepresented in STEM. Examples of positive science voice include these interview responses: *"Tell me about your experience with Skynet Junior Scholars and our after-school astronomy program."* Quin: "*It was fun. When I first came and I wasn't sure though but I stayed. We have to learn about a lot of things like Messier 1 and how the Crab Nebula looks. It's cool and fun."* Interviewer: *"What do you like about it?"* Aran: *"Everything. I'm learning a lot like I never knew, like it takes light years, I didn't even know what light years were."*
- Another key finding is that the participants demonstrated their creativity and individual expressiveness as they created astronomical images.
- Several participants demonstrated significant collaboration through Skype and forum exchanges.

Badges earned by participants in our pilot study.

Activity	First Light	Exposure Time	Scavenger Hunt	Investigate Filters	Make a Color Image	Create it Yourself
Student						
Vivian	X	X	X	X	X	
Jay	X	X	X	X		
Don	X	X	X			
Quin						
Aran	X	X				

Sample Participant Images





Implications

Exposure to science, technology, engineering, and math (STEM) components in SJS point to possible future research on STEM voice. There is limited research related to STEM voice. This project could help define and identify STEM voice. Research on STEM voice has social justice implications. It could reduce alienation students may feel towards STEM, and empower them to cross STEM cultural borders. Developing a STEM voice can help raise academic achievement levels and help close the opportunity gap as students traditionally underrepresented in STEM fields learn to become producers of knowledge instead of just consumers.

References Available Here

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